

Sudhanshu Tiwari

AWS Center for Quantum Computing – Pasadena, CA, USA

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Research Experience

Amazon Web Services, Center for Quantum Computing

Pasadena, CA

Research Scientist II

2024 – Current

- Leading research on scalable architectures for fault-tolerant quantum computing, with a focus on device physics and system-level integration.
- Collaborating with cross-disciplinary teams to translate fundamental research into deployable quantum technologies.
- Mentoring junior researchers and contributing to long-term research strategy within the hardware development group.

Purdue University

West Lafayette, IN

Postdoctoral Research Associate

2021 – 2024

- Principal researcher for DARPA MTO WARP and COFFEE programs, driving experimental design, device development, and program deliverables.
- Advanced the DARPA COFFEE Program to Phase 2 in collaboration with BAE Systems.
- Hosted and presented program milestones to DARPA program managers and industrial collaborators.
- Organized and led weekly research meetings; prepared technical presentations and demonstrations for U.S. government stakeholders.
- Co-authored a successful proposal to the U.S. Army Research Laboratory.
- Mentored graduate and undergraduate researchers in device fabrication, modeling, and experimental methods.

Indian Institute of Science

Bengaluru, India

Graduate Researcher

2015 – 2021

- Conducted independent research in PiezoMEMS, resulting in multiple journal publications and a patent.
- Designed and demonstrated a novel tip-coupled two-cantilever microsystem for viscosity sensing.
- Initiated and sustained collaborations with interdisciplinary teams across materials science, microfabrication, and device physics.

Teaching and Mentorship Experience

Purdue University

West Lafayette, IN

Instructor of Record

2023 – 2024

- Designed and taught a graduate-level course on Microelectromechanical Systems (MEMS).
- Achieved a strong student evaluation score: **4.28/5**.
- Nominated by *four* students for the Postdoctoral Mentor Award in recognition of exceptional guidance and support.

Indian Institute of Science

Bengaluru, India

Teaching Assistant

2018

- Delivered tutorials on COMSOL Multiphysics for MEMS design and supported students through office hours and assignment evaluations.

Education

A.P.J. Abdul Kalam Technical University

Lucknow, India

Bachelor of Technology

2011–2015

Patents

- [1] Connor Devitt, Sunil A Bhawe, and **Sudhanshu Tiwari**. "SINGLE MAGNETIC BIAS MAGNETOSTATIC WAVE FILTER". 2025.
- [2] **Tiwari, Sudhanshu** and Rudra Pratap. "A device to determine viscosity of a fluid and a method thereof". Indian Patent Application Number: 202041024905. 2020.

Journal Publications

- [1] Connor Devitt, **Sudhanshu Tiwari**, Bill Zivasatienraj, and Sunil A Bhawe. "Spin-wave band-pass filters for 6G communication". In: *Nature* (2026).
- [2] **Tiwari, Sudhanshu**, Anuj Ashok, Connor Devitt, Sunil A Bhawe, and Renyuan Wang. "High-performance magnetostatic wave resonators based on deep anisotropic etching of gadolinium gallium garnet substrates". In: *Nature Electronics* (2025).
- [3] Connor Devitt, **Tiwari, Sudhanshu**, Sunil A. Bhawe, and Renyuan Wang. "A Distributed Magnetostatic Resonator". In: *IEEE Transactions on Microwave Theory and Techniques* (2024), pp. 1–8.
- [4] Connor Devitt, Renyuan Wang, **Tiwari, Sudhanshu**, and Sunil A Bhawe. "An edge-coupled magnetostatic bandpass filter". In: *Nature Communications* 15.1 (2024), p. 7764.
- [5] Renyuan Wang, **Tiwari, Sudhanshu***, Yiyang Feng, Sen Dai, and Sunil A Bhawe. "Giant Real-time Strain-Induced Anisotropy Field Tuning in Suspended Yttrium Iron Garnet Thin Films". In: *arXiv preprint arXiv:2405.13303* (2024).
- [6] Yiyang Feng, **Tiwari, Sudhanshu**, Sunil A. Bhawe, and Renyuan Wang. "Micromachined Tunable Magnetostatic Forward Volume Wave Bandstop Filter". In: *IEEE Microwave and Wireless Technology Letters* (2023), pp. 1–4. doi: 10.1109/LMWT.2023.3267449.
- [7] **Tiwari, Sudhanshu**, Ajay Dangi, and Rudra Pratap. "A tip-coupled, two-cantilever, non-resonant microsystem for direct measurement of liquid viscosity". In: *Nature Microsystems & Nanoengineering* 9.1 (2023), p. 34.
- [8] **Tiwari, Sudhanshu**, Randhir Kumar, Ajay Dangi, Jeyaseelan A Antony, Soma Dutta, and Rudra Pratap. "Low cost, contamination-free, and damage-free fabrication of PZT MEMS on SOI substrate". In: *Journal of Micromechanics and Microengineering* 32.2 (2021), p. 025003.
- [9] Randhir Kumar, **Tiwari, Sudhanshu**, Varun Thakur, and Rudra Pratap. "Growth of ultrafast, super dense ZnO nanorods using microwaves for piezoelectric MEMS applications". In: *Materials Chemistry and Physics* 255 (2020), p. 123607.
- [10] Viphretuo Mere, **Tiwari, Sudhanshu**, Aneesh Dash, Rakshitha Kallega, Akshay Naik, Rudra Pratap, and Shankar Kumar Selvaraja. "Photonics Integrated PiezoMEMS-PipMEMS: A Scalable Hybrid Platform for Next-Generation MEMS". In: *IEEE Sensors Letters* 4.12 (2020), pp. 1–4. doi: 10.1109/LENS.2020.3042708.

- [11] **Tiwari, Sudhanshu**, Randhir Kumar, Saurabh Arun Chandorkar, and Rudra Pratap. "Fully Differential Actuation and Sensing in Piezoelectric Diaphragm Resonators for High Signal to Background Resonant Sensing". In: *Journal of Microelectromechanical Systems* 29.5 (2020), pp. 888–893. doi: 10.1109/JMEMS.2020.3001246.
- [12] Christopher Y Cheng, Ajay Dangi, Liqiang Ren, **Tiwari, Sudhanshu**, Robert R Benoit, Yongqiang Qiu, Holly S Lay, Sumit Agrawal, Rudra Pratap, Sri-Rajasekhar Kothapalli, et al. "Thin Film PZT-Based PMUT Arrays for Deterministic Particle Manipulation". In: *IEEE transactions on ultrasonics, ferroelectrics, and frequency control* 66.10 (2019), pp. 1606–1615.
- [13] Ajay Dangi, Christopher Cheng, Sumit Agrawal, **Tiwari, Sudhanshu**, Gaurav Ramesh Datta, Robert Benoit, Rudra Pratap, Susan Trolier-McKinstry, and Sri-Rajasekhar Kothapalli. "A Photoacoustic Imaging Device using Piezoelectric Micromachined Ultrasound Transducers (PMUTs)". In: *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2019).

* Corresponding/Equally Contributing Author

Conference Publications

- [1] Renyuan Wang, Connor Devitt, Eric Langlois, **Tiwari, Sudhanshu**, Anuj Ashok, and Sunil Bhawe. "Temperature Compensated Magnetostatic Wave Resonator Microsystem". In: *2024 IEEE MTT-S International Microwave Symposium*. 2024.
- [2] Javed N, Priyanka Singh, **Tiwari, Sudhanshu**, Pavitra Jain, Kongbrailatpam Sandeep, Rudra Pratap, and Gayathri Pillai. "PZT-based Multi-Mode Cantilever for Viscosity Sensing". In: *2023 Joint Conference of the European Frequency and Time Forum and IEEE International Frequency Control Symposium (EFTF/IFCS)*. 2023, pp. 1–4.
- [3] Vishnu Kumar, **Tiwari, Sudhanshu**, Shreeraj Joshi, and Saurabh Arun Chandorkar. "Novel Estimation Method for Dielectric Loss Quality Factor in Composite SOI/PZT Devices". In: *2022 IEEE 35th International Conference on Micro Electro Mechanical Systems Conference (MEMS)*. 2022, pp. 754–757. doi: 10.1109/MEMS51670.2022.9699467.
- [4] R. Kumar, **Tiwari, Sudhanshu**, and R. Pratap. "Significant Enhancement in Operational Bandwidth of ZnO PMUTs due to the Simultaneous Existence of Softening and Hardening Nonlinearity". In: *2020 IEEE International Ultrasonics Symposium (IUS)*. 2020, pp. 1–4. doi: 10.1109/IUS46767.2020.9251421.
- [5] Viphetuo Mere, **Tiwari, Sudhanshu**, Anish Dash, Rakshita Kallega, Akshay Naik, Rudra Pratap, and Shankar Kumar Selvaraja. "Silicon Photonics Enabled on-Chip Optical Readout of piezoMEMS Resonators". In: *2020 IEEE Sensors*. IEEE. 2020, **Best Paper Award**.
- [6] Ajay Dangi, Sumit Agrawal, **Tiwari, Sudhanshu**, Shubham Jadhav, Christopher Cheng, Gaurav Ramesh Datta, Susan Trolier-McKinstry, Rudra Pratap, and Sri-Rajasekhar Kothapalli. "Ring PMUT array based miniaturized photoacoustic endoscopy device". In: *Photons Plus Ultrasound: Imaging and Sensing 2019*. Vol. 10878. International Society for Optics and Photonics. 2019, p. 1087811.
- [7] **Tiwari, Sudhanshu**, Randhir Kumar, P Krishna Menon, Jeyaseelan A Antony, Soma Dutta, and Rudra Pratap. "On the Realization of Self-Sensing Piezoelectric MEMS Actuators". In: *2019 International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS)*. IEEE. 2019, pp. 1–5.

- [8] Ajay Dangi, Sumit Agrawal, **Tiwari, Sudhanshu**, Shubham Jadhav, Christopher Cheng, Susan Troler-McKinstry, Rudra Pratap, and Sri-Rajasekhar Kothapalli. "Evaluation of High Frequency Piezoelectric Micromachined Ultrasound Transducers for Photoacoustic Imaging". In: *2018 IEEE SENSORS*. IEEE. 2018, pp. 1–4.
- [9] **Tiwari, Sudhanshu**, Randhir Kumar, Ajay Dangi, and Rudra Pratap. "Enabling Fabrication of PZT Based PiezoMEMS Devices". In: *2018 IEEE SENSORS*. IEEE. 2018, pp. 1–4.

Awards

New Employee Recognition Award for Departmental Achievement	<i>ECE Purdue</i>
Nominated by 4 students for the 2024 Postdoctoral Mentor Award	<i>Purdue</i>
Institute Fellowship for Graduate Research	<i>Through GATE Exam</i>
GARP Travel Award	<i>For conference travel from IISc</i>
Travel Grant from IEEE UFFC	<i>For attending $F^2C\pi^2$ 2019 Joint Conference</i>

Services

Reviewer *2020-Current*

- **Nature** Communications
- **IEEE** Sensors Journal
- **IEEE** Microwave and Wireless Technology Letters
- **IOP** Journal of Micromechanics and Microengineering
- **IEEE** Sensors Conference Series

Outreach

- Served as the **Vice-Chair** of the local IEEE chapter at IISc from 2017 to 2018. Secured a competitive grant from the **IEEE Sensors council** for organizing a Summer School on the theme "Nanotechnology: From Science to Systems and Beyond."
- Actively participated in organizing several outreach programs such as **Think Nano** and **Women in Science and Engineering**.
- Coordinator and Instructor of fabrication module for Special Workshop on Piezoelectric Micromachined Ultrasound Transducers for NPOL (DRDO) Scientists.

Grant Writing Experience

Co-authored successful Core Research Grant proposal (SERB, India).
 Contributed to funded ARL proposal on MSW loss-study (Funded).
 Assisted in proposal development for DARPA HOTS program (funded).

Invited Talks

Magnetostatic wave filters: @ IISc, Bengaluru	<i>Dec 2025</i>
MEMS-Engineered Magnetics: The Future of Agile RF Filtering: @ UC Davis, USA	<i>July 2025</i>
Magnetostatic Wave Technology: @ UT El Paso, USA	<i>Feb 2024</i>
Materials to Devices: Piezo MEMS: @ NMII, Bangalore, India	<i>Feb 2018</i>